

Table of Contents

Web-Scale Job Scheduling	1
<i>Walfredo Cirne and Eitan Frachtenberg</i>	
DEMB: Cache-Aware Scheduling for Distributed Query Processing	16
<i>Junyong Lee, Youngmoon Eom, Alan Sussman, and Beomseok Nam</i>	
Employing Checkpoint to Improve Job Scheduling in Large-Scale Systems	36
<i>Shuangcheng Niu, Jidong Zhai, Xiaosong Ma, Mingliang Liu, Yan Zhai, Wenguang Chen, and Weimin Zheng</i>	
Multi-objective Processor-Set Selection for Computational Cluster-Systems	56
<i>N. Peter Drakenberg</i>	
On Workflow Scheduling for End-to-End Performance Optimization in Distributed Network Environments	76
<i>Qishi Wu, Daqing Yun, Xiangyu Lin, Yi Gu, Wuyin Lin, and Yangang Liu</i>	
Dynamic Kernel/Device Mapping Strategies for GPU-Assisted HPC Systems	96
<i>Jiadong Wu, Weiming Shi, and Bo Hong</i>	
Optimal Co-Scheduling to Minimize Makespan on Chip Multiprocessors	114
<i>Kai Tian, Yunlian Jiang, Xipeng Shen, and Weizhen Mao</i>	
Evaluating Scalability and Efficiency of the Resource and Job Management System on Large HPC Clusters	134
<i>Yiannis Georgiou and Matthieu Hautreux</i>	
Partitioned Parallel Job Scheduling for Extreme Scale Computing	157
<i>David Brelsford, George Chochia, Nathan Falk, Kailash Marthi, Ravindra Sure, Norman Bobroff, Liana Fong, and Seetharami Seelam</i>	
High-Resolution Analysis of Parallel Job Workloads	178
<i>David Krakov and Dror G. Feitelson</i>	
Identifying Quick Starters: Towards an Integrated Framework for Efficient Predictions of Queue Waiting Times of Batch Parallel Jobs	196
<i>Rajath Kumar and Sathish Vadhiyar</i>	

On Identifying User Session Boundaries in Parallel Workload Logs 216
 Netanel Zakay and Dror G. Feitelson

Performance and Fairness for Users in Parallel Job Scheduling 235
 Dalibor Klusáček and Hana Rudová

Comprehensive Workload Analysis and Modeling of a Petascale
Supercomputer 253
 Haihang You and Hao Zhang

Author Index 273